

Multi Source Inverter for Electric Vehicles

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Overview (Dorsheed)

- Problem Statement
- Design Process
- Architecture
- Budget & Resource Requirements
- Schematic and PCB Capture
- Evaluation Results & Analysis
- Project Timeline
- Future Work & Conclusion
- Demonstration

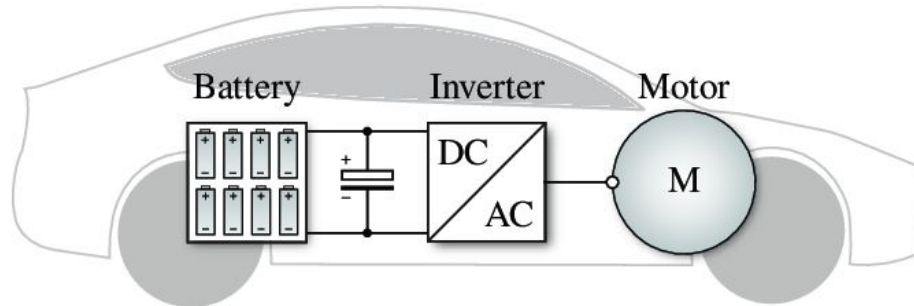
Problem Statement (Dorsheed)

- EV inverters today face limitations in power loss, component count, size, and efficiency, especially under variable load and multi-source operation.
- Existing EV traction inverters typical use a single DC source, which restricts controllability, increases switching losses, and limits adaptability to different operating conditions.
- Modern EV systems require the ability to combine multiple DC sources (e.g., battery cells, PV modules) and dynamically reconfigure them to improve performance and reliability.
- Current MSI topologies are complex, often requiring high switch counts, added control circuitry, and larger form factor. The goal of this project is to reduce component count while retaining full functionality.

Problem Statement (Dorsheed)

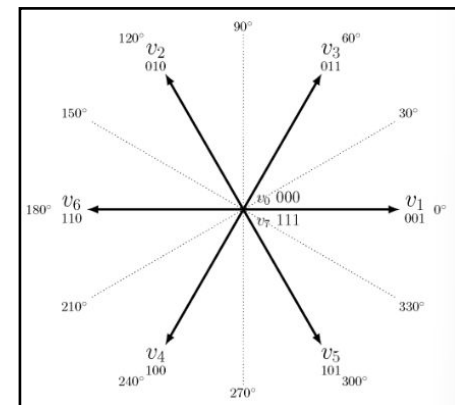
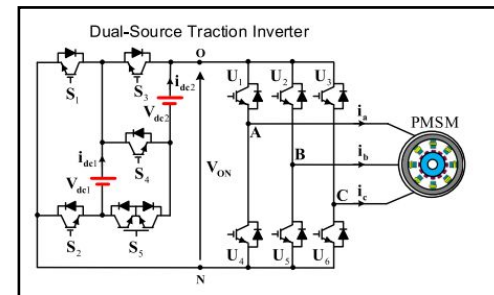
Main design requirements for the project for an efficient MSI:

- Merges multiple DC inputs safely
- Delivers stable three-phase AC output
- Minimize switching losses & harmonics
- Remains feasible for prototype implementation within academic constraints



Current Solutions (Dorsheed)

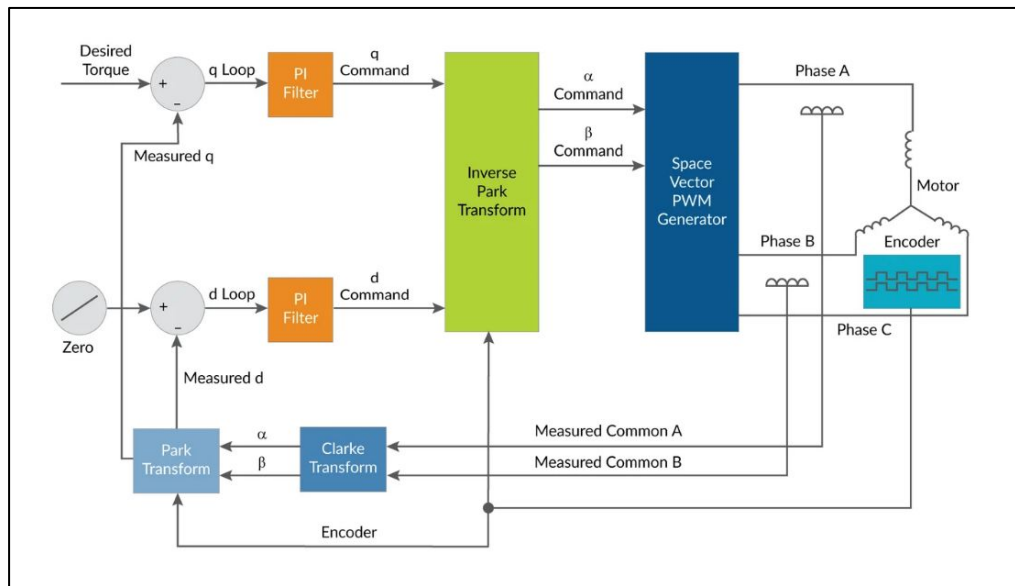
- **Conventional EV traction inverters** use 2-level or 3-level NPC topologies with single DC inputs; they offer good performance but suffer from high switching losses, limited voltage flexibility, and larger component count.
- **PWM/SPWM-controlled inverters** are simple and widely used, but have low DC-bus utilization ($\sim 78\%$) and generate significant harmonics
- **SVPWM-based inverters** improve DC-bus utilization ($>90\%$) and reduce harmonic distortion but still rely on fixed-voltage DC links and single-source architectures.
- **Multi-Source inverter topologies from literature** allow 2-input dynamic voltage combining but require many switches, increasing complexity, size, and cost.
- **Existing source-combination methods** (parallel, series, switching networks) each present drawbacks:
 - **Parallel** → requires identical sources; limited voltage flexibility.
 - **Series** → introduces risk of source imbalance and voltage jumps.
 - **Switched MSI structures** → high control complexity and limited scalability beyond two sources.



Current Solutions

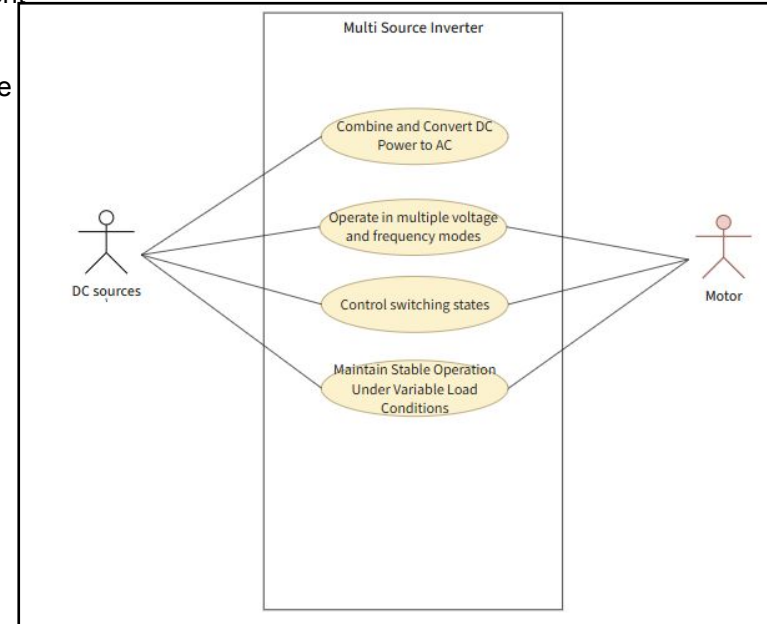
Field Oriented Control (FOC)

- FOC is a closed loop control system which aims to maximize the quadrature force (rotational force) of the motor.
- Measures output currents and motor position to continuously adjusting the switching



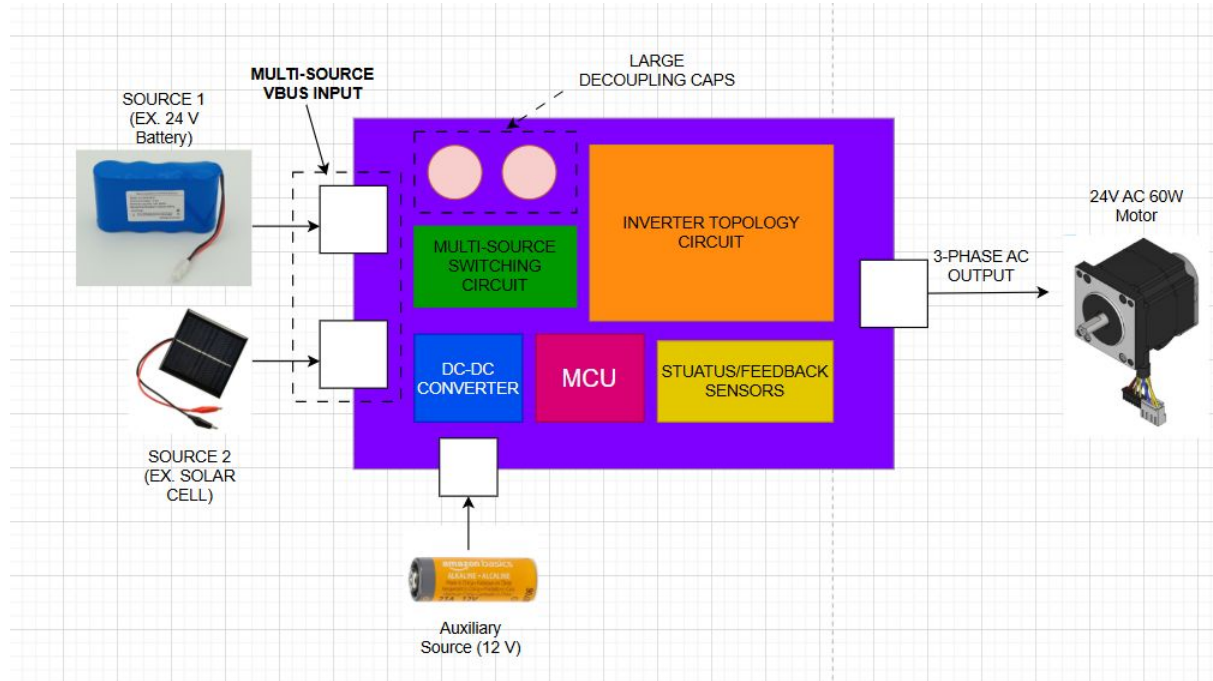
Design Process (Emmanuel)

- **Iterative engineering workflow** based on research → simulation → refinement → hardware prototyping.
- **Requirements gathering & stakeholder alignment** performed early to define performance, size, cost, and compatibility objectives.
- **Concept generation phase** comparing inverter topologies, control methods (PWM/SPWM/SVM), switching devices (IGBT vs MOSFET vs SiC), and source-combination schemes.
- **Field oriented control** is the primary control method used in maximizing the rotational force of the motor
- **PSIM simulation development** to validate switching behavior, losses, controllability, and waveform quality before hardware implementation.
- **Conceptual system design** including subsystem breakdown: power management, filtering, MCU control, gate drivers, switching devices, and AC output.
- **Integration & testing planning** using defined test cases to verify source management, controller behavior, power regulation, safety, and performance.
- **Prototype development phase** to transition from simulation to PCB design, component selection, assembly, and final validation.



Architecture (Dorsheed)

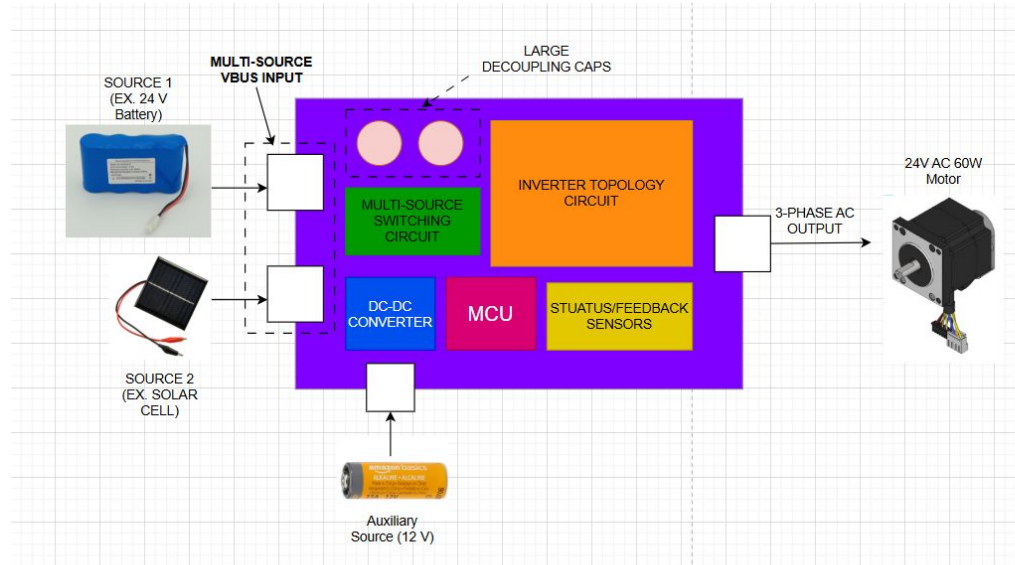
High-Level Design Diagram of MSI board



Architecture (Dorsheed)

Device Subsections

1. Multi-Source Input Section
2. Auxiliary Input Power
3. DC-DC Converter
4. MCU
5. Inverter Topology Circuit
6. Feedback Sensors



Budget & Resource Requirements (Emmanuel)

Materials	Cost per unit(\$)	Quantity	Total Cost (\$)
Motor	109.33	1	109.33
DC Decoupling Caps	~6	2	12
Ceramic Decoupling	~6	1	6
All Resistors	-	-	6
All Diodes	-	-	6
MOSFETs	~10.00	6	60.00
STM32G491VET6 board	35.26	1	35.26
18650 li-ion battery	~17.00	1	17.00
INA240A sensor	3.23	2	6.46

Budget & Resource Requirements (Emmanuel)

Materials	Cost per unit(\$)	Quantity	Total Cost (\$)
Current Shunt	~2.5	1	2.5
Heat sink	~10	1	10
24-12V DCDC	~7.5	1	7.5
24-5V DCDC	~4	1	4
24-3.3V DCDC	~4	1	4
PCB Fabrication	~80	1	80
Cables & Connectors (Molex, wiring, terminal blocks)	~20	-	20
Isolation, Fuses, Fuse Holders	~10	-	10
Debug Hardware	~15	1	15

Budget & Resource Requirements (Emmanuel)

Materials	Cost per unit(\$)	Quantity	Total Cost (\$)
Assembly, Solder, Prototype Expenses	~50	-	50
Total			461.05

Manpower Costs

No. of team members	Hours per week	Salary per hour	No. of weeks in school year	Total (\$)
4	7.25	30.71	24	21,374.16

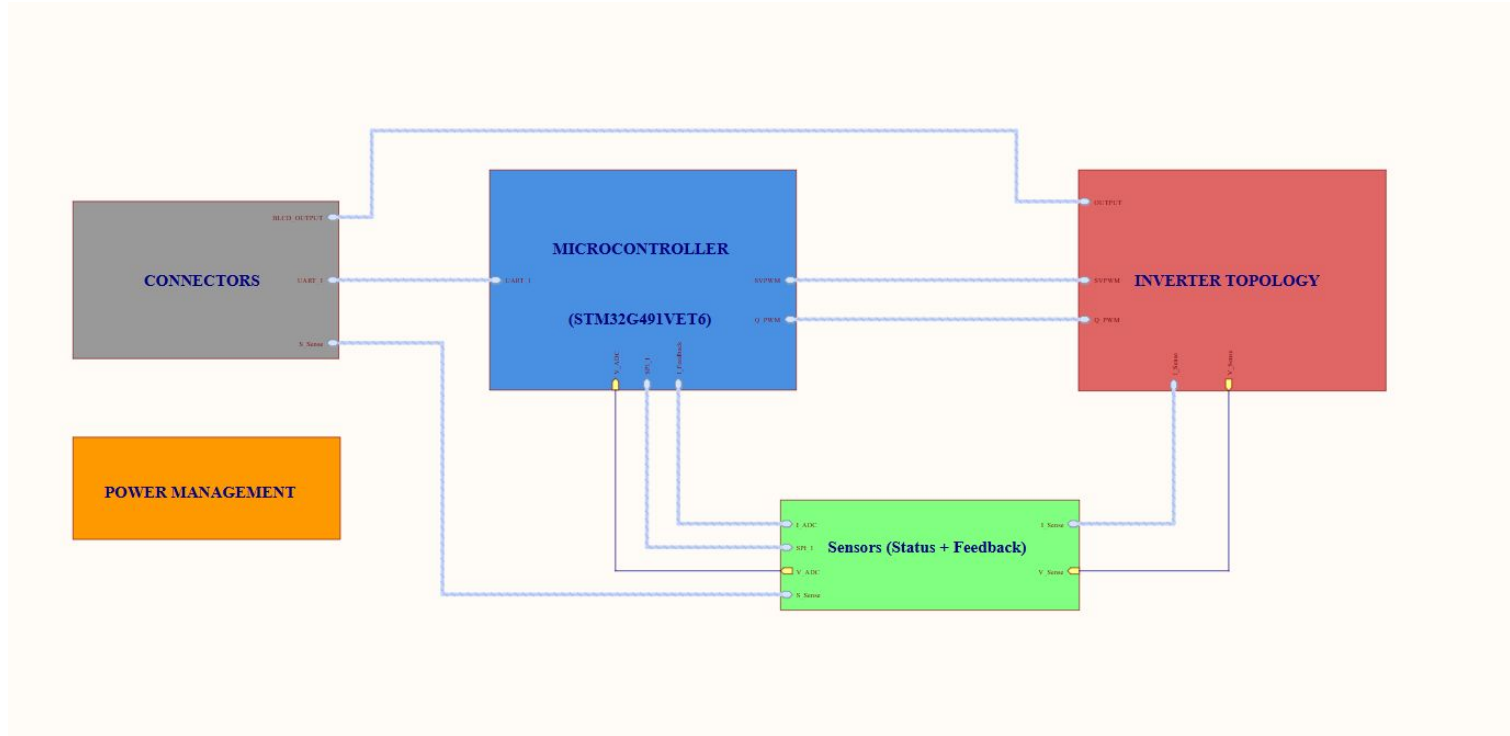
Simulation

Evaluation Results & Analysis

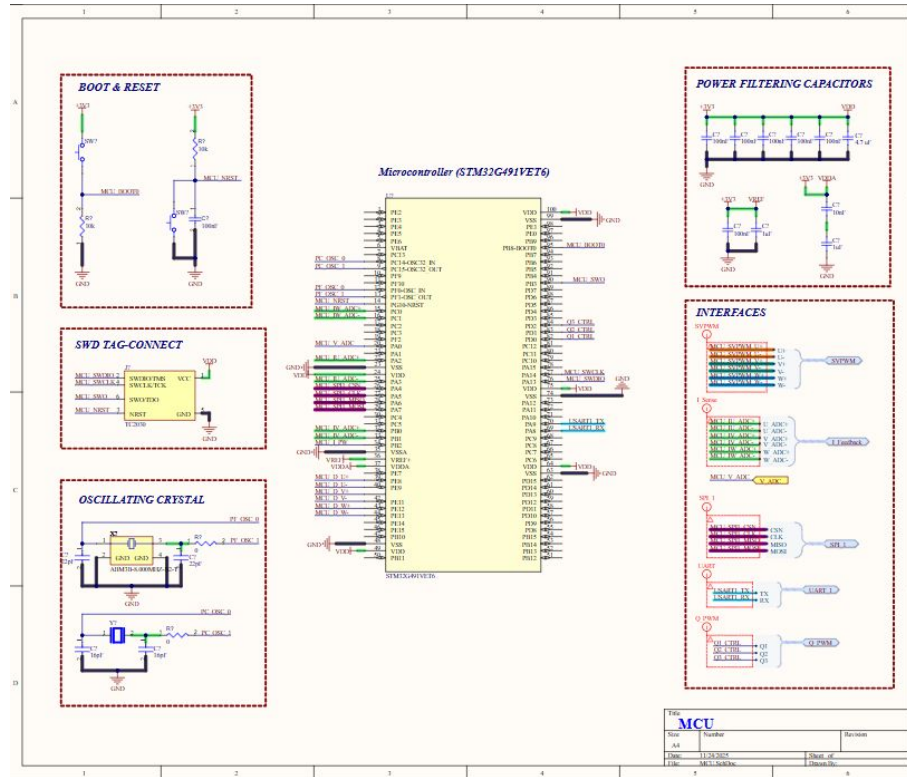
Project Timeline (Badi)

5	Prototyping & Presentation			Nov. 26th 25'
5.1	Hardware prototype development	Nov. 1st 25'	Nov. 18th 25'	
5.2	Team Presentation & Demo	Nov. 19th 25'	Nov. 26th 25'	
6	Team Retrospective Report			Dec. 2nd 25'
6.1	Develop Report	Nov. 22nd 25'	Dec. 2nd 25'	
7	Preparations for prototype			Jan. 5th 26'
7.1	Place orders on components and meet with advisor for PCB design analysis	Dec. 2nd 25'	Jan. 5th 26'	
8	Prototype Development			N/A
8.1	Build prototype	Week 1	Week 4	
8.2	Conduct Integration testing	Week 3	Week 6	

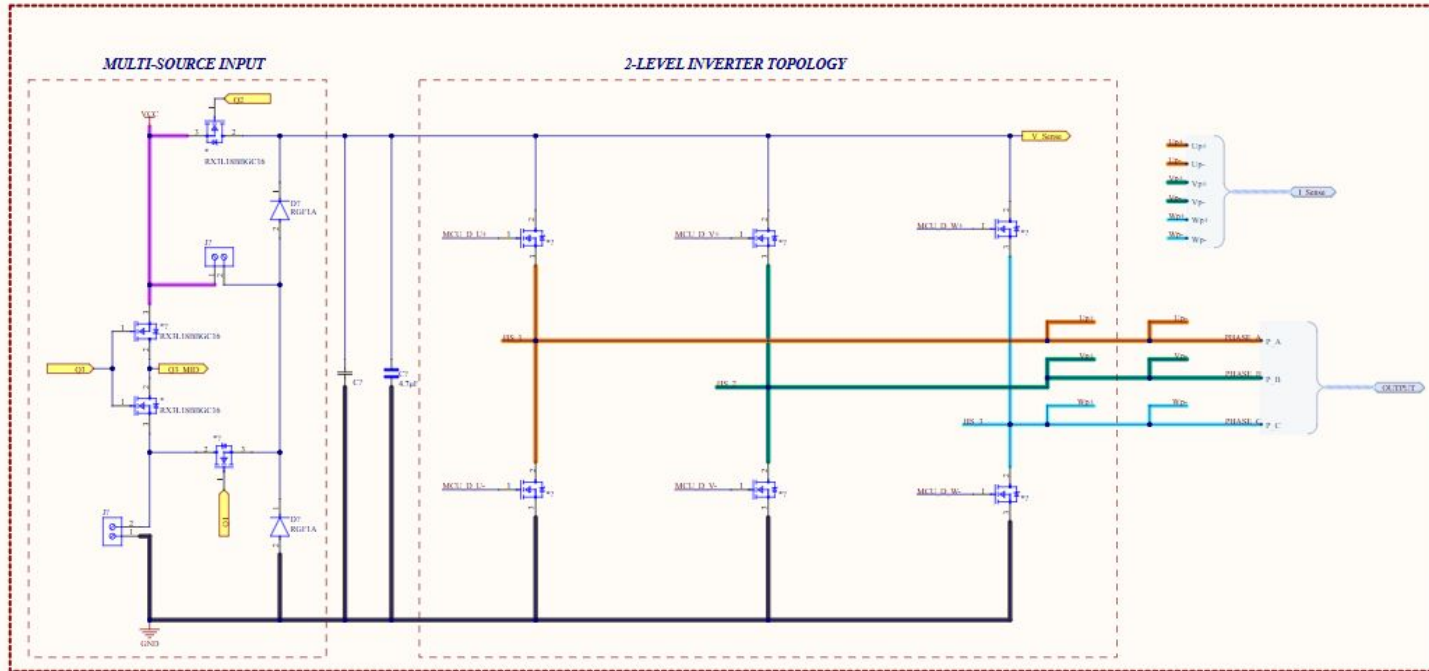
Schematic Showcase - Block Section



Schematic Showcase - Microcontroller Section



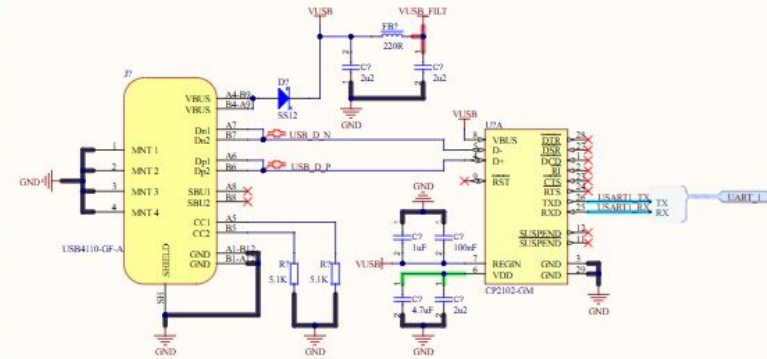
Schematic Showcase - Inverter Topology Section



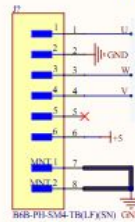
Q5 = Back-to-Back MOSFET
Half-bridge Switch

Schematic Showcase - Connectors Section

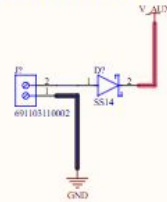
USB-C CONNECTOR + USB-TO-UART



AC-SIDE CONNECTORS

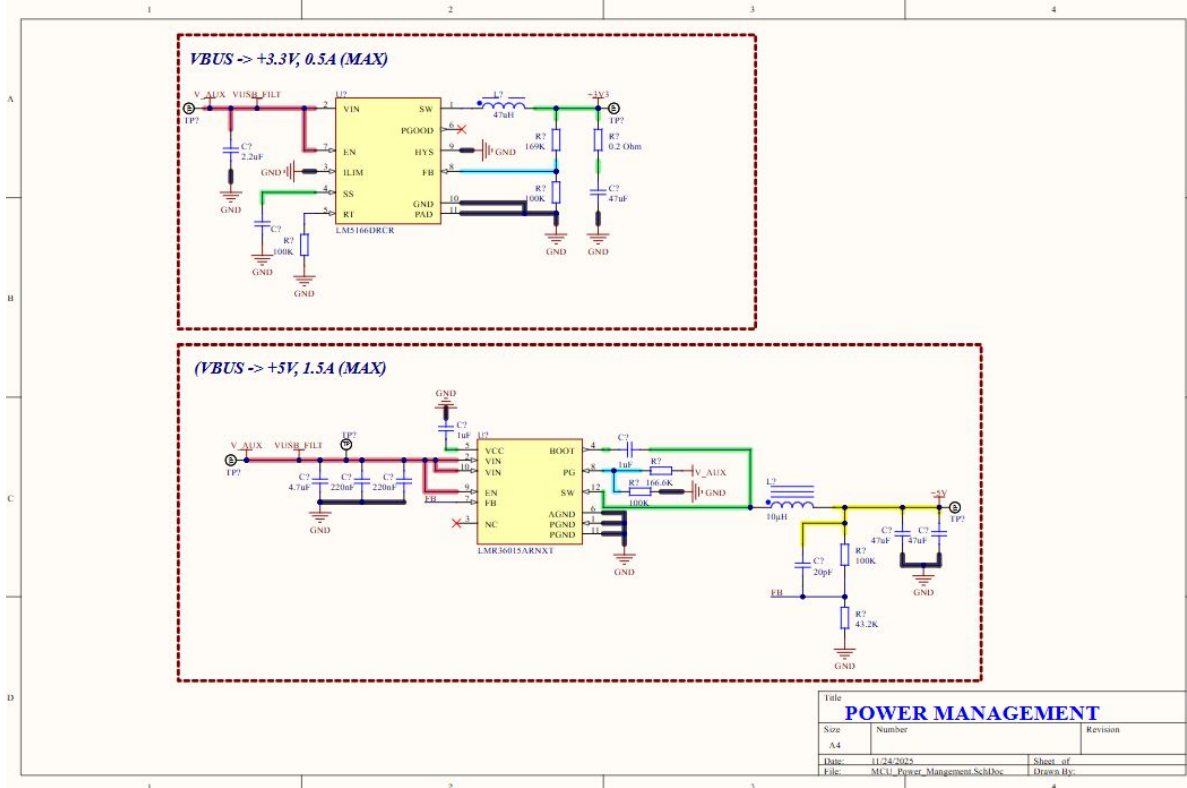


DC-SIDE AUXILIARY CONNECTOR

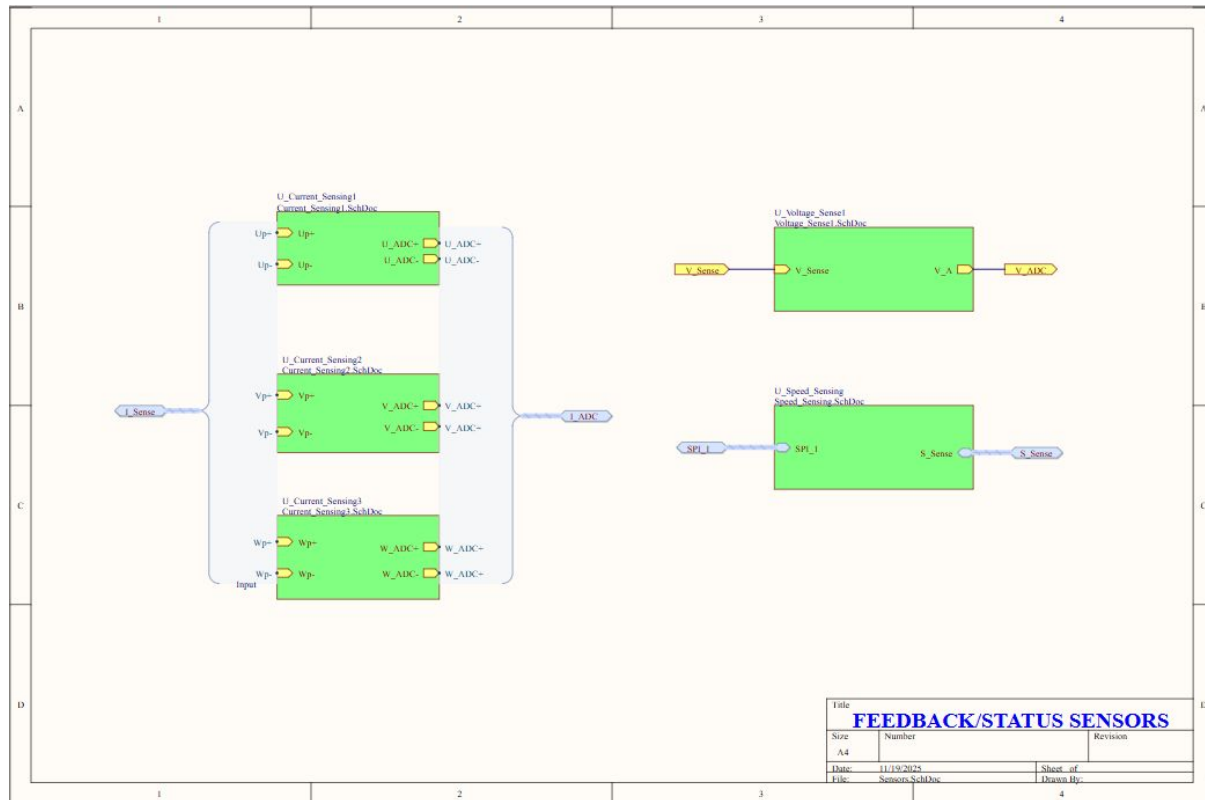


CONNECTORS		
Size	Number	Revision
A4		
Date:	11/24/2021	Sheet of
File:	Connector.SchDoc	Drawn By:

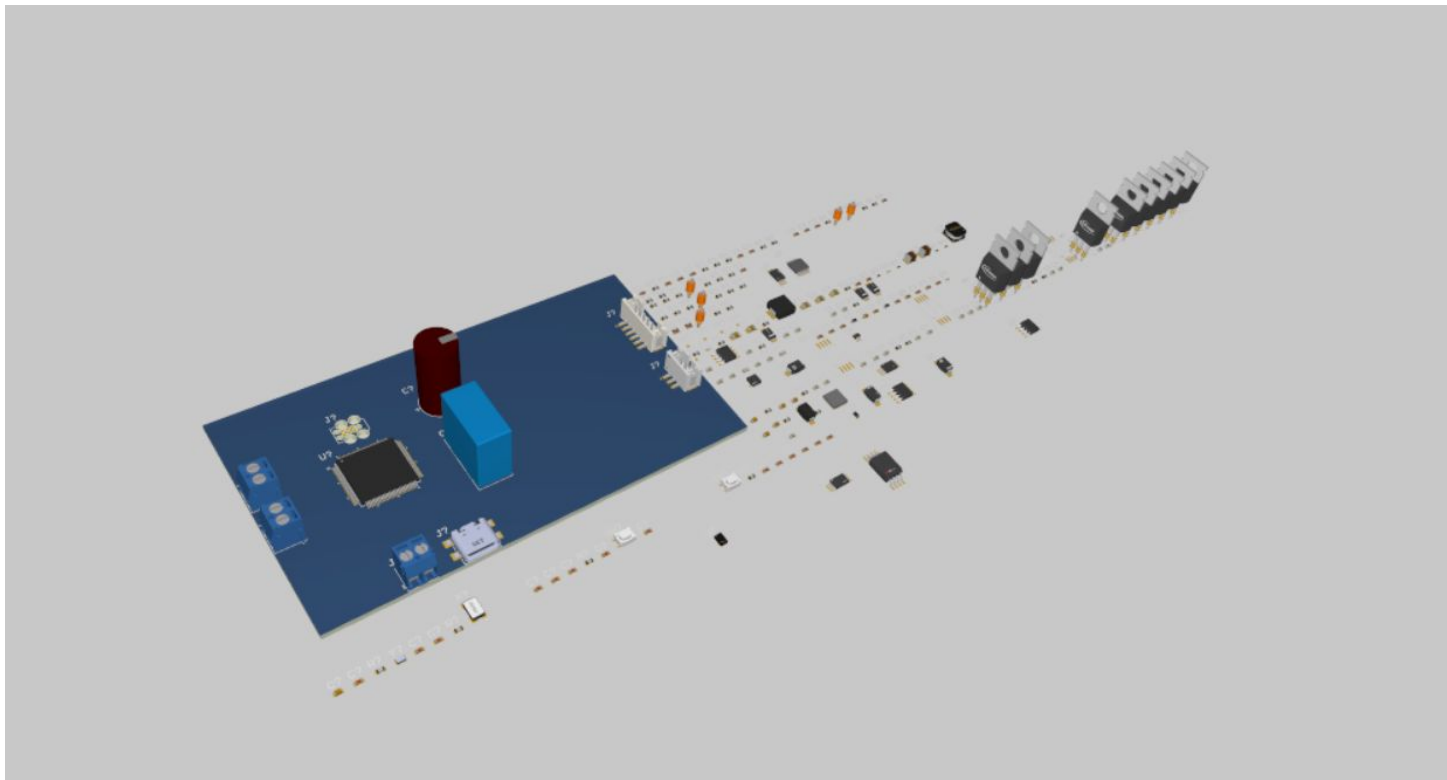
Schematic Showcase: DC-DC Converter Section



Schematic Showcase: Sensor Circuit Section



3D PCB View Showcase



Future Work & Conclusion (Dorsheed-Badi)

Q&A